

Happy Supplies Parts Warehouse

a) Assumptions:

- Each customer has a unique CustomerNumber.
- Each employee may help multiple customers.
- Each customer order may contain multiple parts.
- Each part has a unique PartNumber.
- Each part has exactly one Name, Type, CageCode, and UnitPrice.
- CageCode identifies where a part is stored.
- CustomerType depends only on CustomerNumber.
- QuantityOrdered depends on both the order and the part.
- Date and Time belong to the order, not individual parts.

b) It is already in 1NF, except for the customer's name.

CustomerNumber
CustomerFirstName
CustomerLastName
CustomerType
Date
Time
Employee
PartNumber
PartName
Type
CageCode
QuantityOrdered
UnitPrice

Step1: 2NF- no partial dependency, to do that, we split into tables.

customer	part	order	order_line
CustomerNumber (PK)	PartNumber (PK)	CustomerNumber (PK, FK)	CustomerNumber (PK, FK)
CustomerFirstName	PartName	Date (PK)	Date (PK, FK)
CustomerLastName	Type	Time (PK)	Time (PK, FK)
CustomerType	CageCode	Employee	PartNumber (PK, FK)
	UnitPrice		QuantityOrdered

Step2: 3NF- no transitive dependencies. The table above does not have any transitive dependencies, so it is already in 3rd normal form.

c) **1NF**: (CustomerNumber, Date, Time, PartNumber)

2NF: CUSTOMER: CustomerNumber, PART: PartNumber, ORDER: (CustomerNumber, Date, Time), ORDER_LINE: (CustomerNumber, Date, Time, PartNumber)

3NF: same as 2NF (no further decomposition required)

Foreign Keys in 3NF: CustomerNumber in ORDER → CUSTOMER, (CustomerNumber, Date, Time) in ORDER_LINE → ORDER, PartNumber in ORDER_LINE → PART

d) **customer**(CustomerNumber (PK), CustomerName, CustomerType)
part(PartNumber (PK), PartName, Type, CageCode, UnitPrice)
order(CustomerNumber (PK, FK), Date (PK), Time (PK), Employee)
order_line(CustomerNumber (PK, FK), Date (PK, FK), Time (PK, FK), PartNumber (PK, FK), QuantityOrdered)

Therapist

- a) Assumptions
- staffNo uniquely identifies a therapist.
 - patNo uniquely identifies a patient.
 - therapistName depends only on staffNo.

- patName depends only on patNo.
- A therapist may work at multiple branches.
- On any given date/time, a therapist sees one patient at one branch.
- A patient can have multiple appointments per day.

b) This table is already in 1NF, except for first and last names, and appointment dates.

staffNo	therapistName	patNo	patName	appointment date	time	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

Step1- 2NF- We remove partial dependencies: therapist names depend only on staffNo, patient names depend only on patNo. So we separate them.

therapist	patient	appointment
staffNo (PK)	patNo (PK)	staffNo (PK, FK)
firstName	firstName	patNo (FK)
lastName	lastName	appointmentDate (PK)
		appointmentTime (PK)
		branchNo

Step2- 3NF there is no transitive dependencies, so it is already in 3rd normal form.

- c) **1NF:** (staffNo, appointmentDate, appointmentTime)
2NF: THERAPIST: staffNo, PATIENT: patNo, APPOINTMENT: (staffNo, appointmentDate, appointmentTime)
3NF: same as 2NF (no further decomposition required)
 Foreign Keys in 3NF: staffNo in APPOINTMENT -> THERAPIST, patNo in APPOINTMENT -> PATIENT

d) **therapist**(staffNo (PK), therapistFirstName, therapistLastName)

patient(patNo (PK), patFirstName, patLastName)

appointment(staffNo (PK, FK), appointmentDate (PK), appointmentTime (PK),
patNo (FK), branchNo)

Maid Better

a) Assumptions

- eNo uniquely identifies an employee.
- contractNo uniquely identifies a contract.
- Each contract applies to exactly one event.
- An event has one location.
- An employee may work multiple contracts.
- A contract may have multiple employees.
- hours depends on both eNo and contractNo.

b)

eNo	contractNo	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

It is already in 1NF.

Step 1: 2NF

employee
eNo (PK)
eName

contract
contractNo (PK)
eventNo
eventLoc

employee_contract
eNo (PK, FK)
contractNo (PK, FK)
hours

Step 2: 3NF remove transitive dependency. We have to remove contractNo->eventNo->eventLoc

employee	event	contract	employee_contract
eNo (PK)	eventNo (PK)	contractNo (PK)	eNo (PK, FK)
eName	eventLoc	eventNo	contractNo (PK, FK)
			hours

c) **1NF**: (eNo, contractNo)

2NF: EMPLOYEE: eNo, CONTRACT: contractNo, EMPLOYEE_CONTRACT: (eNo, contractNo)

3NF: EMPLOYEE: eNo, EVENT: eventNo, CONTRACT: contractNo, EMPLOYEE_CONTRACT: (eNo, contractNo)

Foreign Keys in 3NF:

eventNo in CONTRACT → EVENT,

eNo in EMPLOYEE_CONTRACT → EMPLOYEE,

contractNo in EMPLOYEE_CONTRACT → CONTRACT

d) **employee**(eNo (PK), eName)

event(eventNo (PK), eventLoc)

contract(contractNo (PK), eventNo (FK))

employee_contract(eNo (PK, FK), contractNo (PK, FK), hours)